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45. (i) Half reaction are

$\overset{+7}{M}nO_4^- \longrightarrow Mn^{2+}$ ..... (1)

and $Sn^{2+} \longrightarrow Sn^{4+}$ ..... (2)

(1) $\Rightarrow$ $MnO_4^- + 8H^- + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

(2) $\Rightarrow$ $Sn^{2+} \longrightarrow Sn^{4+} + 2e^-$

(1) $\times$ 2 $\Rightarrow$ $2MnO_4^- + 16H^- + \cancel{10e^-} \longrightarrow 2Mn^{2+} + 8H_2O$

(2) $\times$ 5 $\Rightarrow$ $5\,Sn^{2+} \longrightarrow 5Sn^{4+} + \cancel{10e^-}$

$\Rightarrow$ $2MnO_4^- + 5Sn^{2+} + 16H^+ \longrightarrow 2Mn^{2+} + 5Sn^{4+} + 8H_2O$

(ii)

$\overset{+3}{C_2}O_4^{2-} \longrightarrow \overset{+4}{C}O_2$ ..... (1)

$\overset{+6}{Cr_2}O_7^{2-} \longrightarrow Cr^{3+}$ ..... (2)

(1) $\Rightarrow$ $C_2O_4^{2-} \longrightarrow 2\,CO_2 + 2\,e^-$ ..... (3)

$Cr_2O_7^{2-} + 14\,H^+ \longrightarrow 2\,Cr^{3+} + 7\,H_2O$ ..... (4)

(3) $\times$ 3 $\Rightarrow$ $3\,C_2O_4^{2-} \longrightarrow 6\,CO_2 + \cancel{6\,e^-}$ ..... (5)

$Cr_2O_7^{2-} + 14\,H^+ + 6\,e^- \longrightarrow 2\,Cr^{3+} + 7\,H^2O$ ..... (4)

$Cr_2O_7^{2-} + 3\,C_2O_4^{2-} + 14\,H^+ \longrightarrow 2\,Cr^{3+} + 6\,CO_2 + 7\,H_2O$

(iii)

half reaction $\Rightarrow$ $S_2O_3^{2-} \longrightarrow S_4O_6^{2-}$ ..... (1)

$I_2 \longrightarrow I^-$ ..... (2)

(1) $\Rightarrow$ $2S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + \cancel{2e^-}$ ..... (3)

(2) $\Rightarrow$ $I_2 + \cancel{2e^-} \longrightarrow 2I^-$ ..... (4)

(3) + (4) $\Rightarrow$ $2S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2-} + 2I^-$

(iv) Half reactions are

$\overset{0}{Z}n \longrightarrow Zn^{2+}$ ..... (1)

$\overset{+5}{N}O_3^- \longrightarrow \overset{+2}{N}O$ ..... (2)